

Course: B. Tech. Branch : Information Technology Semester : VII

Subject Code & Name: Machine Learning (BTITOE704B)

Max Marks: 60

Date: 08/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain sampling theory along with its types.	L1	6
B) Explain cross validation along with its various types.	L1	6
C) Explain the hypothesis testing with respect to a machine learning system with example.	L2	6
Q.2 Solve Any Two of the following.		12
A) Demonstrate Naive Bayes Classifier.	L2	6
B) Explain PAC learnability.	L3	6
C) Explain Bias-variance tradeoff with diagram.	L2	6
Q. 3 Solve Any Two of the following.		12
A) Demonstrate multilayer perceptron model.	L2	6
B) Explain Gradient descent in detail.	L2	6
C) Explain Back-propagation neural network with its architecture.	L2	6
Q.4 Solve Any Two of the following.		12
A) Write a short note on autoencoders	L3	6
B) Explain various regularization techniques in machine learning.	L2	6
C) Write a short note on ensemble modelling technique.	L3	6
Q. 5 Solve Any Two of the following.		12
A) Explain vanishing and exploding gradients problems in detail.	L3	6
B) Write a short note on guided back propagation.	L2	6
C) Write a short note on recurrent neural network.	L2	6

*** End ***